

Department of Electronics and Communication Engineering
CUIET, Chitkara University Punjab**Course Name:** IoT Application Development**Course Code:** 23EC362**Batch:** 2023**Credits:** 4**Session:** Jan-Jun 2026**Semester:** 6th**Date:** 16 Feb 2026**Class/Group:** IoT Vertical

Lab Evaluation – 1

Course Outcome Covered	List of Experiments Evaluated R – Remembering, U – Understanding, P – Applying, N – Analyzing, E – Evaluating, C - Creating	Evaluation Component / marks
CO1, CO5	1. To configure a GPIO pin as digital output and implement LED blinking to illustrate basic edge-node operation 2. To interface a push button as digital input with an LED as digital output for simple sensor–actuator control. 3. To configure the USART peripheral for asynchronous serial transmission of text data to a terminal.	PBL implementation = 15 (U, P, N) Practical File = 5 (R, U) Total Marks = 20

Problem Statement 1: Smart Traffic Signal Module (Smart City / Transportation Domain)

In modern cities, traffic lights operate intelligently to ensure smooth vehicle movement and minimize congestion. As a part of a smart transportation ecosystem, modular edge controllers manage individual signal lights and communicate their state with the central traffic management console.

As an embedded systems engineer, you are required to design and develop a Smart Traffic Signal Module on the STM32L475 IoT Node.

Specifications:

- Configure one GPIO pin as digital output to drive an LED representing a traffic signal (RED).
- Implement a blinking pattern to simulate warning or transition mode — LED ON for 500 ms, OFF for 500 ms, continuously.
- Configure the USART peripheral to periodically transmit a message (e.g., “SIGNAL_ACTIVE”) to a PC terminal for remote monitoring.
- The system should start and stop blinking only when specific user input (button press) is received.

Project Outcome:

A deployable prototype simulating an intelligent traffic light node capable of visual signaling and serial communication with a central controller.

Project Domain: Embedded Systems / Smart Transportation

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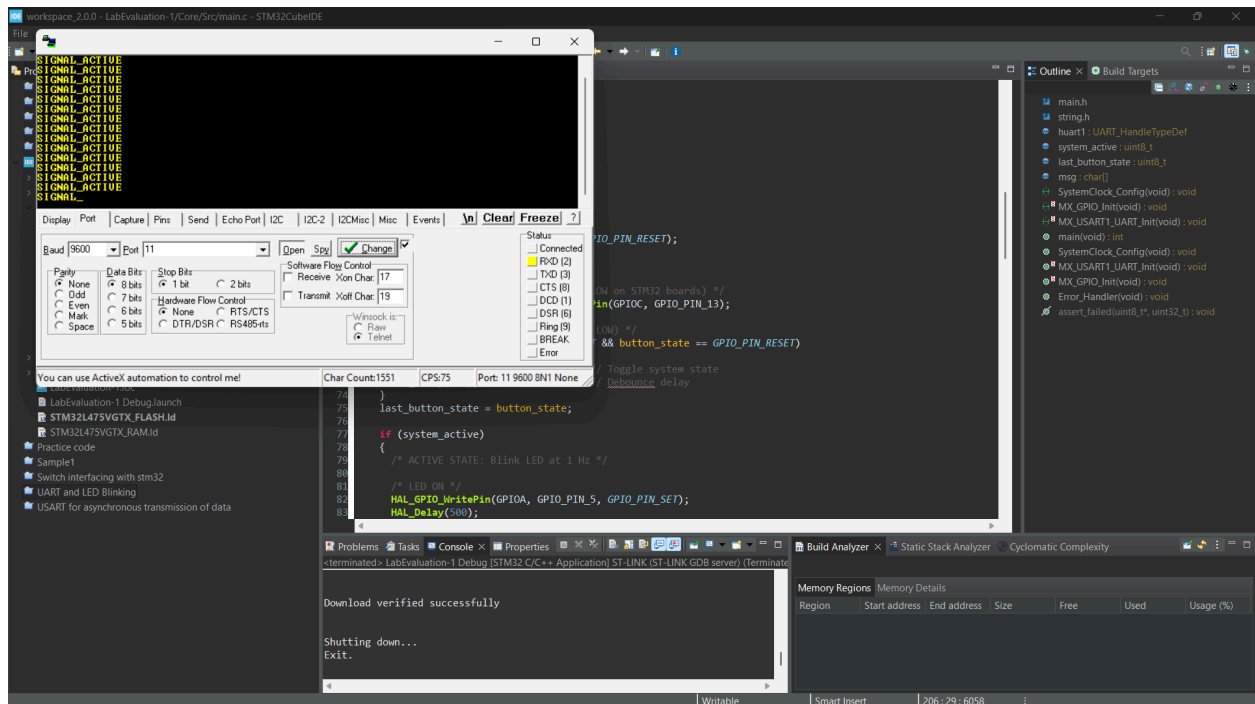
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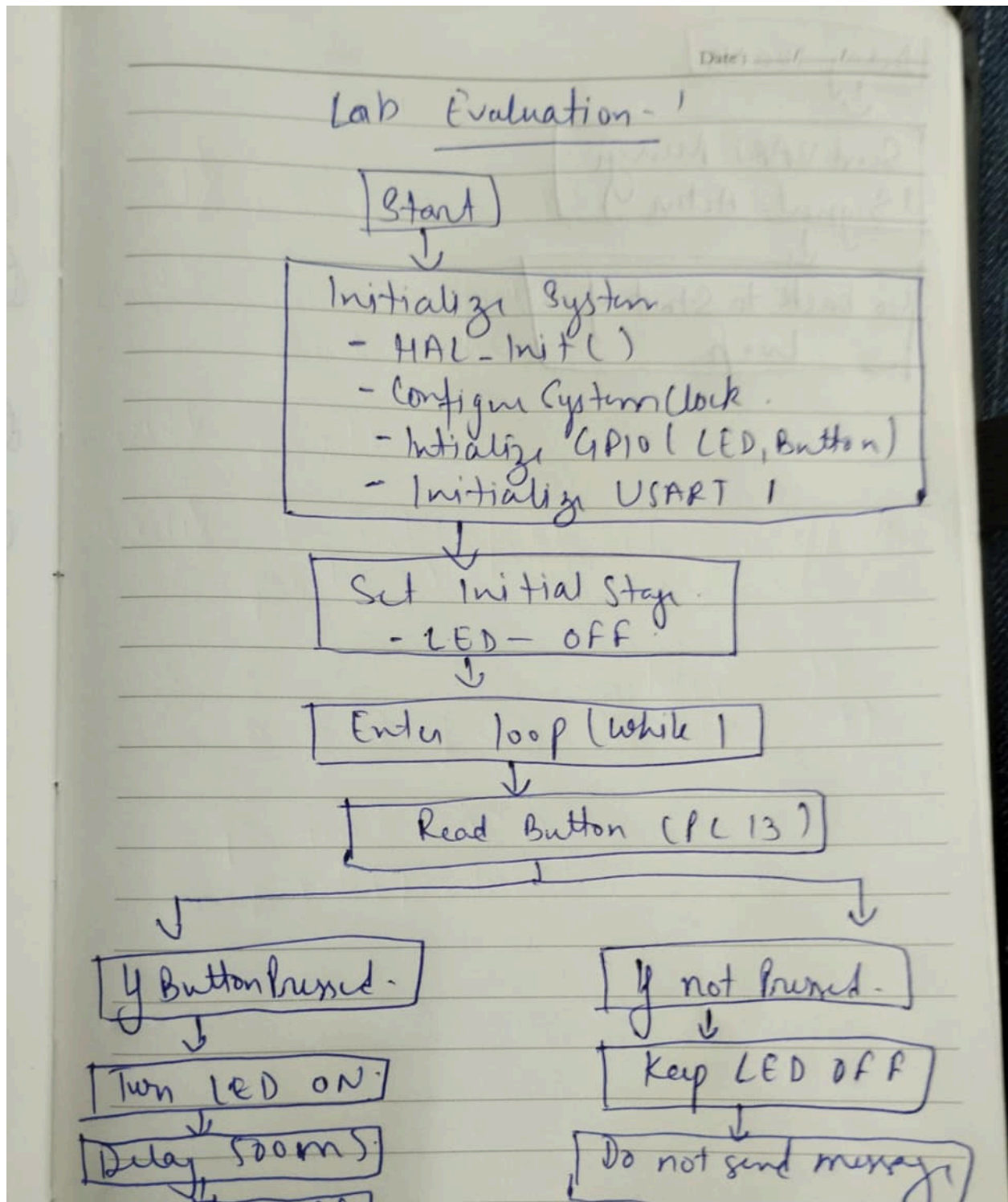
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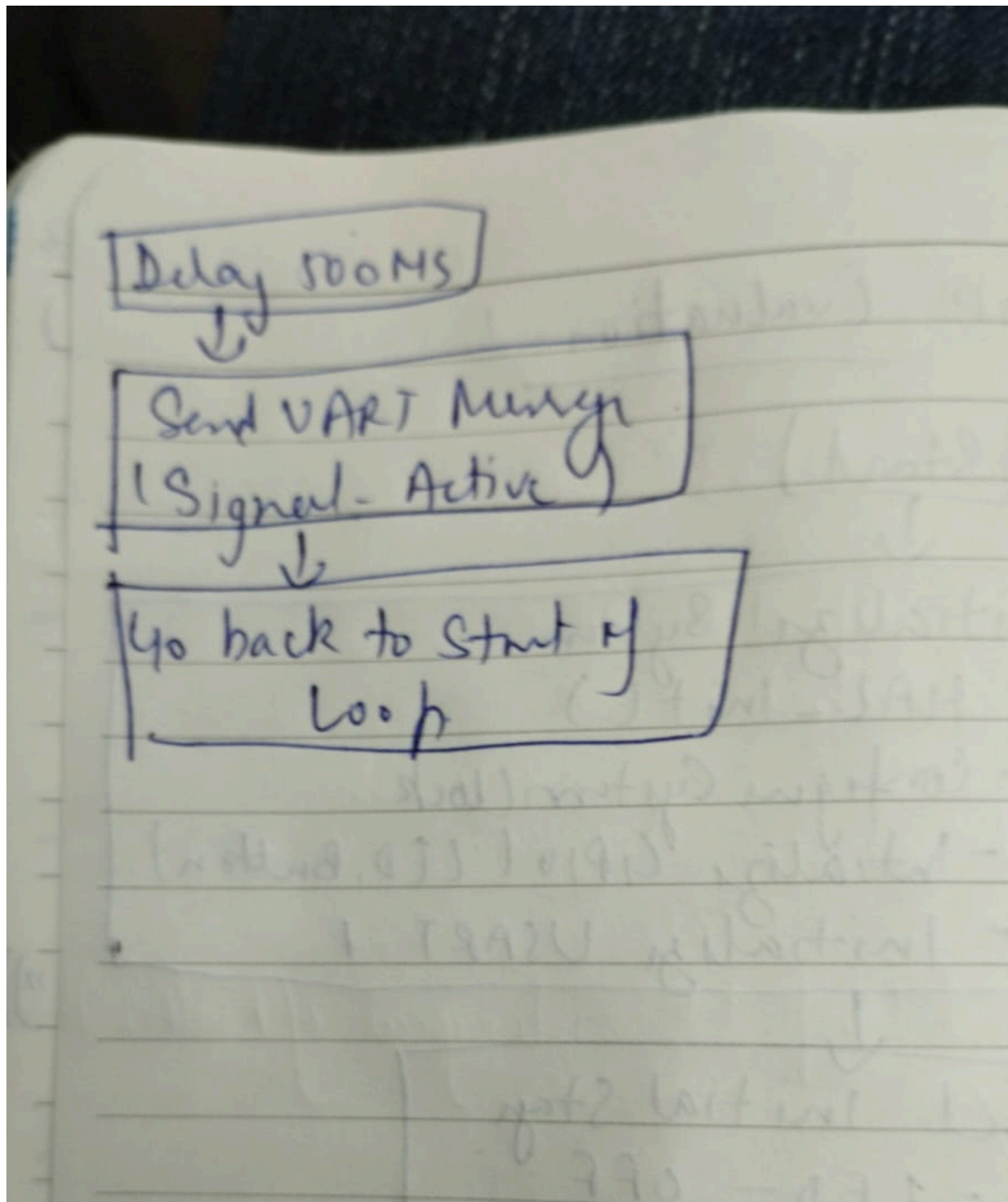
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After completing this PBL, I learned how to configure and use GPIO and USART peripherals on the STM32 using CubeMX and CubeIDE. I understood how to interface a push button as an input and an LED as an output and control them using HAL functions. I also learned how to generate a blinking pattern using delays and how to send periodic status messages over UART for monitoring. Another important learning was implementing simple control logic using a state variable to switch between IDLE and ACTIVE modes. Overall, this project helped me understand the practical working of an embedded system as a small edge device in a smart application.

Evaluation Rubric:

Criterion/Marks	5	3	1
Problem Understanding (Flowchart/system Design)	Complete, well-labeled flowchart showing peripheral init → main loop → input decision → output control → UART reporting. Clear symbols, arrows, decisions properly marked.	Basic flowchart present with main blocks (init, loop, input/output). Some labels missing or flow unclear at decision points.	Incomplete flowchart or hand-drawn sketch without proper symbols/logic flow. Major gaps in understanding shown.
PBL Implementation (Code & Hardware)	Code compiles/runs perfectly. All peripherals (GPIO, UART, timer) correctly configured via HAL/CubeMX. Modular structure, proper comments, hardware wiring correct.	Code works with minor issues. Basic peripheral config correct but may have inefficient loops or missing error checks. Hardware functional.	Code has compilation errors or major logic flaws. Peripherals misconfigured. Hardware connections faulty or incomplete.
Demonstration	Full functionality demonstrated — LED behavior perfect, UART messages exactly as specified, confident explanation of working.	Core functionality works (LED responds, UART transmits) but timing/messages have minor issues. Basic explanation provided.	Demo partially works or fails frequently. Cannot explain working properly or show expected outputs.



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Practical File	File is complete in all aspects, observation and reflection summary is correct	File is complete in all aspects, observation and reflection summary has some minor issues	File is complete but poorly maintained, reflection summary is not written well
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Course Coordinator Name and Signature